PROBLEM 5 | Problems T2_CrystStruct.pdf

Crystalline Structures & Miller Indices

# Planar and Linear Densities and Mass of an Aluminium Bar

Original statement · Problems T2_CrystStruct.pdf

*5. Given that aluminum has an FCC structure, the atomic radius* $1.43\ \text{Å}$ *and the atomic weight* $26.98\ \text{g/mol}$*, calculate: a) Planar density, in* $\text{at/cm}^2$*, for the* $(110)$ *plane. (Solution:* $\rho_{(110)} = 8.7 \times 10^{14}\ \text{at/cm}^2$*) b) Linear density, in* $\text{at/cm}$*, for the* $[110]$ *direction. (Solution:* $\rho_{[110]} = 3.5 \times 10^{7}\ \text{at/cm}$*) c) Weight of a bar of* $20\ \text{mm}$ *in diameter and* $1\ \text{m}$ *long. (Solution:* $m = 855\ \text{g}$ *) Data:* $1\ \text{m} = 10^{10}\ \text{Å}$*;* $N_A = 6.023 \times 10^{23}\ \text{mol}^{-1}$*.*

**Solution:** $\rho_{(110)} = 8.7 \times 10^{14}\ \text{at/cm}^2$

PHASE 1: HYPOTHESIS & PARAMETERS

### Input Data:

- Structure: Face-Centred Cubic (FCC) $\implies n = 4$ atoms/cell.
- Atomic radius: $R = 1.43\ \text{Å} = 1.43 \times 10^{-8}\ \text{cm} = 0.143\ \text{nm}$.
- Molar atomic weight: $M = 26.98\ \text{g/mol}$.
- Aluminium bar dimensions:
- Diameter: $D = 20\ \text{mm} = 2.0\ \text{cm}$.
- Length: $L = 1\ \text{m} = 100\ \text{cm}$.
- Avogadro constant: $N_A = 6.023 \times 10^{23}\ \text{mol}^{-1}$.

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PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

1. **Lattice Parameter in FCC [Session 3 Slide 27]:**

The atoms touch along the face diagonal:

$$4R = a\sqrt{2} \implies a = 2\sqrt{2}R$$

2. **Planar Density on the** $(110)$ **Plane [Session 4 Slide 9]:**

The $(110)$ plane is a rectangle of base $a\sqrt{2}$ and height $a$. It contains 2 net atoms:

$$\rho_{(110)} = \frac{2}{\sqrt{2}a^2} = \frac{\sqrt{2}}{a^2}$$

3. **Linear Density in the $[110]$ Direction [Session 4 Slide 8]:**

The $[110]$ direction is the face diagonal, with length $L_{[110]} = a\sqrt{2} = 4R$. It contains 2 net atoms:

$$\rho_{[110]} = \frac{2}{4R} = \frac{1}{2R}$$

4. **Mass of the Cylindrical Bar:**

From the theoretical volumetric density:

$$\rho_v = \frac{n \cdot M}{a^3 \cdot N_A}$$

The volume of the cylindrical bar is:

$$V = \pi \left(\frac{D}{2}\right)^2 L$$

The total mass is $m = \rho_v \cdot V$.

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`PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP`

### 1. Lattice Parameter ($a$):

$$a = 2\sqrt{2}R = 2 \times \sqrt{2} \times (1.43 \times 10^{-8}\ \text{cm}) = 4.04465 \times 10^{-8}\ \text{cm} = 4.045\ \text{Å}$$

### 2. Part a: Planar Density on $(110)$:

$$A_{(110)} = \sqrt{2}a^2 = \sqrt{2} \times (4.04465 \times 10^{-8}\ \text{cm})^2 = \sqrt{2} \times 1.6359 \times 10^{-15}\ \text{cm}^2 = 2.3135 \times 10^{-15}\ \text{cm}^2$$

The plane contains $N_{\text{atoms}} = 4 \times \frac{1}{4} + 2 \times \frac{1}{2} = 2$ atoms:

$$\rho_{(110)} = \frac{2 \text{ at}}{2.3135 \times 10^{-15} \text{ cm}^2} = 8.645 \times 10^{14} \approx \mathbf{8.6 \times 10^{14}} \text{ at/cm}^\mathbf{2}$$

**Discrepancy with the official solution.** The official key gives $8.7 \times 10^{14}$ at/cm$^2$; the given radius ($R = 1.43$ Å) yields $8.645 \times 10^{14}$ at/cm$^2$, i.e. $8.6 \times 10^{14}$ to two significant figures. The official figure is reproduced only if $a$ is first rounded to $4.04$ Å ($8.66 \times 10^{14}$); that rounding is not applied here.

### 3. Part b: Linear Density on $[110]$:

Since $[110]$ is the close-packed direction ($L = 4R$):

$$\rho_{[110]} = \frac{1}{2R} = \frac{1}{2 \times (1.43 \times 10^{-8} \text{ cm})} = \frac{1}{2.86 \times 10^{-8} \text{ cm}} = 3.4965 \times 10^7 \approx \mathbf{3.5 \times 10^7} \text{ at/cm}$$

### 4. Part c: Mass of the Bar:

- **Unit cell volume:**

$$V_C = a^3 = (4.04465 \times 10^{-8} \text{ cm})^3 = 6.6167 \times 10^{-23} \text{ cm}^3$$

- **Volumetric density of aluminium:**

$$\rho_v = \frac{4 \times 26.98 \text{ g/mol}}{(6.6167 \times 10^{-23} \text{ cm}^3) \times (6.023 \times 10^{23} \text{ mol}^{-1})} = \frac{107.92}{39.853} = 2.708 \text{ g/cm}^3$$

- **Macroscopic volume of the bar:**

$$V = \pi \left( \frac{2.0 \text{ cm}}{2} \right)^2 \times (100 \text{ cm}) = 100\pi \approx 314.159 \text{ cm}^3$$

- **Calculated total mass:**

$$m = \rho_v \cdot V = (2.708 \text{ g/cm}^3) \times (314.159 \text{ cm}^3) = \mathbf{850.7} \text{ g} \approx \mathbf{851} \text{ g}$$

**Discrepancy with the official solution.** The official key gives $m = 855$ g. The given data ($R = 1.43$ Å, $M = 26.98$ g/mol, $N_A = 6.023 \times 10^{23}$ mol$^{-1}$) give $\rho = 2.708$ g/cm$^3$ and $m = 850.7$ g ($\approx 851$ g, $0.5\%$ below the key). 855 g would require $\rho \approx 2.72$ g/cm$^3$, which is inconsistent with the computed $2.708$ g/cm$^3$ and is not used.

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PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

- **Consistency with Aeronautical Aluminium:** The calculated theoretical density of $2.708\ \mathrm{g/cm^3}$ reproduces the reference value for light aerospace aluminium alloys (such as the 2024-T3 and 7075-T6 series, with $\rho \approx 2.7$–$2.8\ \mathrm{g/cm^3}$), whose low specific weight is the backbone of pressurised fuselage and wing-skin airframe structures.